



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# LESSON PLAN

For



## **Automate Video and Voice AI Agents with n8n**

TGS Ref No: TGS-2024052081

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version v3.0**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                                                                                                                                               | Author        |
|----------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| v1.0           | 8 July 2026               | Initial release.                                                                                                                                                                                                                                          | Dr Alfred Ang |
| v2.0           | 11 July 2026              | Agentic AI loop engineering rewrite; labs 1-6.                                                                                                                                                                                                            | Dr Alfred Ang |
| v3.0           | 12 July 2026              | Aligned 100% to the runnable labs: RAG concepts, Retell + ngrok setup, knowledge base and voice cloning, Vapi FAQ agent, Wav2Lip/MuseTalk/HeyGen lip-sync comparison, interactive avatars (in-browser and LiveAvatar), and Gemini Veo 3 video generation. | Dr Alfred Ang |

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## Course profile

|                        |                                              |
|------------------------|----------------------------------------------|
| <b>Provider</b>        | <b>Tertiary Infotech Academy Pte Ltd</b>     |
| <b>Course code</b>     | TGS-2024052081                               |
| <b>Version</b>         | v3.0 (12 July 2026)                          |
| <b>Mode</b>            | Instructor-led adult training, hands-on labs |
| <b>Duration</b>        | 5 training days, 8 instructional hours each  |
| <b>Practical ratio</b> | At least 70 percent hands-on lab time        |
| <b>Trainer</b>         | Dr Alfred Ang                                |

## Trainer strategy

Demonstrate each concept with one small working example, then move quickly into learner practice. For every lab, ask learners to show the evidence - the n8n execution trace, the generated media, the call transcript, the scorecard. Do not accept a screenshot of a finished screen as proof that the workflow is correct: a fluent output and a correct workflow are not the same thing, and telling them apart is the skill this course teaches.

## Topic map

| <b>Session</b> | <b>Topic</b>                                          | <b>Focus</b>                                                                                                                                                                                  | <b>Labs</b>                        |
|----------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>1</b>       | Topic 01 - Foundations of Agentic AI Loop Engineering | Set up the local stack and learn the engineering loop used in every lab: define the task, give the agent tools, observe behavior, evaluate outputs, improve the workflow, and add guardrails. | Lab 1.1, Lab 1.2, Lab 1.3          |
| <b>2</b>       | Topic 02 - Local AI Agents and RAG with n8n           | Build local agents with Ollama, memory, retrieval, chunking, and document-                                                                                                                    | Lab 2.1, Lab 2.2, Lab 2.3, Lab 2.4 |

|          |                                                                     |                                                                                                                                                                 |                                                                        |
|----------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
|          |                                                                     | grounded answers that can be tested and improved.                                                                                                               |                                                                        |
| <b>3</b> | Topic 03 - Customer Experience and Tool-Using Agents                | Turn retrieval into a workplace assistant that can answer customer questions, collect structured data, and call workflow tools.                                 | Lab 3.1, Lab 3.2, Lab 3.3, Lab 3.4                                     |
| <b>4</b> | Topic 04 - Voice AI Agents                                          | Design, connect, test, and improve a voice booking agent using Retell, n8n webhooks, and a browser front end.                                                   | Lab 4.1, Lab 4.2, Lab 4.3, Lab 4.4, Lab 4.5, Lab 4.6, Lab 4.7, Lab 4.8 |
| <b>5</b> | Topic 05 - AI Video and Avatar Automation                           | Generate scripts, avatar videos, text-to-video clips with Veo 3.1, and open-source talking-head video pipelines that run entirely on the learner's own machine. | Lab 5.1, Lab 5.2, Lab 5.3, Lab 5.4, Lab 5.5                            |
| <b>6</b> | Topic 06 - Publishing, Interactive Avatars, and Capstone Operations | Publish video outputs, deploy interactive avatars, monitor automations, and assemble a production-ready AI workforce capstone.                                  | Lab 6.1, Lab 6.2, Lab 6.3, Lab 6.4, Lab 6.5                            |

## Daily schedule

Total taught content: 32 hours 30 minutes across 29 labs.

### Day 1

| <b>Time</b>              | <b>Duration</b> | <b>Topic / Activity</b>                              | <b>Deck</b> | <b>Evidence produced</b>                                                              |
|--------------------------|-----------------|------------------------------------------------------|-------------|---------------------------------------------------------------------------------------|
| <b>9:30am - 10:30am</b>  | 60 min          | Lab 1.1 - Set Up the Local AI Automation Workstation | Slide 13    | A checked workstation screenshot plus a short note explaining the local architecture. |
| <b>10:30am - 11:15am</b> | 45 min          | Lab 1.2 - Map the Agentic AI Loop Before Building    | Slide 19    | An agentic loop canvas ready to guide the remaining labs.                             |
| <b>11:15am - 12:00pm</b> | 45 min          | Lab 1.3 - Create a Workflow Quality Baseline         | Slide 24    | A baseline quality checklist plus a saved echo workflow checkpoint.                   |
| <b>12:00pm - 1:00pm</b>  | 60 min          | Lab 2.1 - Build Your First Local AI Agent            | Slide 35    | A working local AI agent with a documented system message.                            |
| <b>1:00pm - 2:00pm</b>   | 60 min          | LUNCH                                                | -           | -                                                                                     |
| <b>2:00pm - 2:45pm</b>   | 45 min          | Lab 2.2 - Add Memory and Session Design              | Slide 41    | A memory-enabled agent and a privacy note.                                            |
| <b>2:45pm - 4:00pm</b>   | 75 min          | Lab 2.3 - Build a PDF RAG IT Support Agent           | Slide 46    | A RAG chatbot that answers from the IT FAQ and refuses unrelated questions.           |
| <b>4:00pm - 5:00pm</b>   | 60 min          | Lab 2.4 - Improve RAG with Chunking and Evaluation   | Slide 53    | A RAG evaluation sheet and selected chunking configuration.                           |
| <b>5:00pm - 6:15pm</b>   | 75 min          | Lab 3.1 - Build a Course Advisory CX Agent           | Slide 59    | A working course advisory chatbot embedded in the sample                              |

|                        |        |                                                                                                                                                                                                |   |                                        |
|------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------|
|                        |        |                                                                                                                                                                                                |   | website.                               |
| <b>6:15pm - 6:30pm</b> | 15 min | Concept delivery, review and evidence check - the trainer teaches the concept, workflow and quality slides for the day's labs, then learners show their executions, transcripts and scorecards | - | Trainer sign-off on the day's evidence |

Day 1 instructional time: 8h 00m (9:30am-6:30pm, less a 1-hour lunch; tea breaks are taken inside the sessions).

## Day 2

| <b>Time</b>              | <b>Duration</b> | <b>Topic / Activity</b>                               | <b>Deck</b> | <b>Evidence produced</b>                                                   |
|--------------------------|-----------------|-------------------------------------------------------|-------------|----------------------------------------------------------------------------|
| <b>9:30am - 10:30am</b>  | 60 min          | Lab 3.2 - Add Structured Lead Capture                 | Slide 67    | A structured lead capture branch ready for CRM or spreadsheet integration. |
| <b>10:30am - 11:30am</b> | 60 min          | Lab 3.3 - Create a Tool-Calling Booking Request Agent | Slide 72    | A safe mock booking request workflow.                                      |
| <b>11:30am - 12:30pm</b> | 60 min          | Lab 3.4 - Add Safety Guardrails and Escalation        | Slide 77    | A guardrail test set and passing CX workflow.                              |
| <b>12:30pm - 1:30pm</b>  | 60 min          | LUNCH                                                 | -           | -                                                                          |
| <b>1:30pm - 2:30pm</b>   | 60 min          | Lab 4.1 - Design a Voice Agent Conversation           | Slide 86    | A tested voice conversation design script.                                 |

|                        |        |                                                                                                                                                                                                |           |                                                      |
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| <b>2:30pm - 3:45pm</b> | 75 min | Lab 4.2 - Connect Retell Web Calls Through n8n                                                                                                                                                 | Slide 93  | A secure browser-to-Retell call flow.                |
| <b>3:45pm - 4:45pm</b> | 60 min | Lab 4.3 - QA the Voice Agent with Call Analytics                                                                                                                                               | Slide 103 | A voice QA scorecard and one improved voice prompt.  |
| <b>4:45pm - 5:45pm</b> | 60 min | Lab 4.4 - Add Human Handoff and Notifications                                                                                                                                                  | Slide 108 | A staff handoff branch for bookings and escalations. |
| <b>5:45pm - 6:30pm</b> | 45 min | Concept delivery, review and evidence check - the trainer teaches the concept, workflow and quality slides for the day's labs, then learners show their executions, transcripts and scorecards | -         | Trainer sign-off on the day's evidence               |

Day 2 instructional time: 8h 00m (9:30am-6:30pm, less a 1-hour lunch; tea breaks are taken inside the sessions).

### Day 3

| <b>Time</b>             | <b>Duration</b> | <b>Topic / Activity</b>                                       | <b>Deck</b> | <b>Evidence produced</b>                                                                                             |
|-------------------------|-----------------|---------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------|
| <b>9:30am - 10:30am</b> | 60 min          | Lab 4.5 - Ground the Voice Agent with a Retell Knowledge Base | Slide 113   | A Retell Knowledge Base attached to the voice agent, plus a transcript showing three grounded answers and one honest |

|                          |        |                                                                                                                          |           |                                                                                                                           |
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|                          |        |                                                                                                                          |           | refusal.                                                                                                                  |
| <b>10:30am - 11:30am</b> | 60 min | Lab 4.6 - Clone Your Own Voice and Give It to the Agent                                                                  | Slide 122 | A cloned voice used by the working voice agent, plus a short written note on consent and disclosure.                      |
| <b>11:30am - 12:45pm</b> | 75 min | Lab 4.7 - Book the Appointment into Google Calendar                                                                      | Slide 131 | A voice booking that produces a real, correctly-timed Google Calendar event.                                              |
| <b>12:45pm - 1:45pm</b>  | 60 min | LUNCH                                                                                                                    | -         | -                                                                                                                         |
| <b>1:45pm - 3:00pm</b>   | 75 min | Lab 4.8 - Build a FAQ Voice Agent with Vapi                                                                              | Slide 139 | A working Vapi FAQ voice agent whose brain is an n8n workflow, proven by a curl test, a live call and one honest refusal. |
| <b>3:00pm - 4:15pm</b>   | 75 min | Lab 5.1 - Lip-Sync Face-Off: Wav2Lip vs MuseTalk vs HeyGen                                                               | Slide 152 | A side-by-side MuseTalk/Hey Gen render of one gemma4 script, a scored comparison, and a written recommendation.           |
| <b>4:15pm - 5:15pm</b>   | 60 min | Lab 5.2 - Build a Video Script Agent                                                                                     | Slide 161 | A timed script and shot list ready for video generation.                                                                  |
| <b>5:15pm - 6:30pm</b>   | 75 min | Concept delivery, review and evidence check - the trainer teaches the concept, workflow and quality slides for the day's | -         | Trainer sign-off on the day's evidence                                                                                    |

|  |  |                                                                       |  |  |
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|  |  | labs, then learners show their executions, transcripts and scorecards |  |  |
|--|--|-----------------------------------------------------------------------|--|--|

Day 3 instructional time: 8h 00m (9:30am-6:30pm, less a 1-hour lunch; tea breaks are taken inside the sessions).

## Day 4

| Time              | Duration | Topic / Activity                                                  | Deck      | Evidence produced                                                                                                      |
|-------------------|----------|-------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------|
| 9:30am - 10:45am  | 75 min   | Lab 5.3 - Generate an Avatar News Video with HeyGen               | Slide 166 | A generated avatar news video and quality review notes.                                                                |
| 10:45am - 12:00pm | 75 min   | Lab 5.4 - Generate a Cinematic Video with Veo 3.1 and Gemini      | Slide 172 | A generated Veo clip, the gemma4 shot script that produced it, and a note on what changed between two prompt versions. |
| 12:00pm - 1:30pm  | 90 min   | Lab 5.5 - Build the Free Local Avatar Video Pipeline              | Slide 179 | A free local avatar video and comparison notes.                                                                        |
| 1:30pm - 2:30pm   | 60 min   | LUNCH                                                             | -         | -                                                                                                                      |
| 2:30pm - 3:45pm   | 75 min   | Lab 6.1 - Publish the Avatar Video to YouTube                     | Slide 185 | An unlisted YouTube upload test with approved metadata.                                                                |
| 3:45pm - 5:00pm   | 75 min   | Lab 6.2 - Build an Interactive Avatar That Renders in the Browser | Slide 190 | A working browser avatar, the measured latency, and a                                                                  |

|                        |        |                                                                                                                                                                                                |           |                                                                                                                    |
|------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------|
|                        |        |                                                                                                                                                                                                |           | note naming the three settings that produced it.                                                                   |
| <b>5:00pm - 6:15pm</b> | 75 min | Lab 6.3 - Embed an Interactive HeyGen Avatar with LiveAvatar                                                                                                                                   | Slide 197 | A working streaming avatar session plus the two-column scorecard comparing it against the browser-rendered avatar. |
| <b>6:15pm - 6:30pm</b> | 15 min | Concept delivery, review and evidence check - the trainer teaches the concept, workflow and quality slides for the day's labs, then learners show their executions, transcripts and scorecards | -         | Trainer sign-off on the day's evidence                                                                             |

Day 4 instructional time: 8h 00m (9:30am-6:30pm, less a 1-hour lunch; tea breaks are taken inside the sessions).

## Day 5

| <b>Time</b>              | <b>Duration</b> | <b>Topic / Activity</b>                            | <b>Deck</b> | <b>Evidence produced</b>                          |
|--------------------------|-----------------|----------------------------------------------------|-------------|---------------------------------------------------|
| <b>9:30am - 10:30am</b>  | 60 min          | Lab 6.4 - Monitor, Debug, and Recover AI Workflows | Slide 204   | A monitoring runbook and one tested error branch. |
| <b>10:30am - 12:30pm</b> | 120 min         | Lab 6.5 - Capstone -                               | Slide 209   | A capstone package                                |

|                         |         |                                                                                                                                                                                                |   |                                                        |
|-------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------|
|                         |         | Build a Human-AI Workforce Automation                                                                                                                                                          |   | suitable for trainer assessment and learner portfolio. |
| <b>12:30pm - 1:30pm</b> | 60 min  | LUNCH                                                                                                                                                                                          | - | -                                                      |
| <b>1:30pm - 4:15pm</b>  | 165 min | Concept delivery, review and evidence check - the trainer teaches the concept, workflow and quality slides for the day's labs, then learners show their executions, transcripts and scorecards | - | Trainer sign-off on the day's evidence                 |
| <b>4:15pm - 4:30pm</b>  | 15 min  | Assessment briefing - instruments, timing, and what evidence is graded                                                                                                                         | - | Completed answer script submitted on the LMS           |
| <b>4:30pm - 5:30pm</b>  | 60 min  | Written Assessment (SAQ) - 6 open-ended knowledge questions (K1-K6)                                                                                                                            | - | Completed answer script submitted on the LMS           |
| <b>5:30pm - 6:30pm</b>  | 60 min  | Case Study (CS) - 6 scenario tasks drawn from the labs (A1-A6)                                                                                                                                 | - | Completed answer script submitted on the LMS           |

Day 5 instructional time: 8h 00m (9:30am-6:30pm, less a 1-hour lunch; tea breaks are taken inside the sessions).

## Assessment

| Instrument                                    | Duration | What it tests                                                        | When                                    |
|-----------------------------------------------|----------|----------------------------------------------------------------------|-----------------------------------------|
| <b>Briefing for Assessment</b>                | 15 min   | The learner knows the instruments, the criteria and the appeal route | Before the assessment, on the final day |
| <b>Written Assessment (WA) - Short Answer</b> | 60 min   | KNOWLEDGE: the concepts behind the labs                              | Final day, after the briefing           |
| <b>Case Study (CS)</b>                        | 60 min   | APPLICATION: judgement on a workplace scenario built from the labs   | Final day, after the WA                 |

Both instruments are open book, individually attempted, and marked Competent / Not Yet Competent. A learner assessed NYC is re-assessed once, on the failed instrument only.

## Evidence the trainer collects

- Workflow exports and n8n execution traces for each lab.
- The RAG evaluation sheet (Lab 2.4) with two chunking configurations scored.
- Voice call transcripts and the QA scorecard (Lab 4.3).
- The lip-sync comparison scorecard: Wav2Lip vs MuseTalk vs HeyGen on one script (Lab 5.1).
- Generated media: the avatar news video, the Veo clip, the interactive avatar session.
- The capstone workflow export and its one-page operating guide.